REGRESSION ANALYSIS (LECTURE 1)

DR. KOUSHIK GHOSH

Department of Mathematics, University Institute of Technology, The University of Burdwan, Golapbag (North), Burdwan-713104, India

E-mail: koushikg123@yahoo.co.uk/koushikg123@gmail.com

Consider a Real Life Example

	Item 1	Item 2	Item 3	Item 4	Item 5
Price in Delhi (in Rs):	30	20	70	50	80
Price in Kolkata (in Rs):	25	16	65	48	76

Find the price of a new item in Kolkata having price Rs 90 in Delhi. Also find the price of a new item in Delhi having price Rs 20 in Kolkata.

Introduction to Regression:

The lexical (dictionary) meaning of the word 'Regression' is 'Stepping Back' or 'Going Back'. From the point of view of Statistics 'Regression' is the measure of average relationship between two or more variables in terms of the original units of the corresponding data. By means of regression one can make an attempt to understand the nature of relationship between the variables in order to make a prediction or forecasting.

Given a set of observations for a certain number of variables for a particular population. The statistical technique of estimating the unknown value of one variable (i.e. the dependent variable) from the known value (s) of the other variable (s) [i.e. independent variable (s)] using this set of observations is coined as 'Regression Analysis'.

A Brief History of Regression Analysis:

The earliest form of Regression was the ‘Method of Least Squares’ published by Legendre in the year 1805 and by Gauss in 1809.

The term ‘regression’ was first used in Statistics by British Biometrician Sir Francis Galton. He first introduced the concept of ‘regression to the mean’. Galton first noticed it in connection with his genetic study of the size of seeds, but it is perhaps his 1886 study of human height (height of adult children to the heights of their parents) that really caught the attention of the academicians.

Importance of Regression Analysis:

- Regression studies the nature of relation between two or more than two variables so that by knowing the value of one variable or more than one variables one can predict the corresponding value of remaining variable.
- Regression clearly expresses the cause and effect relationship between the independent and dependent variables. The independent variable (s) is/are the cause (es) and the dependent variable is the effect.

Difference between Correlation and Regression:

- Correlation is a measure of degree or intensity of linear relationship between two variables. Regression studies the nature of relation between two variables so that by knowing the value of one variable one can predict the corresponding value of other variable.
- Correlation cannot provide the causal (cause and effect) relationship between two variables. Regression clearly expresses the cause and effect relationship between two variables. The independent variable is the cause and the dependent variable is the effect.
- Correlation does not help in making prediction. In regression we can make the prediction using the regression line.

Difference between Correlation and Regression (Contd.):

- Correlation coefficients are symmetrical i.e. between two variables X and Y we always have $r_{X,Y} = r_{Y,X}$. But regression coefficients are not generally symmetrical i.e. between two variables X and Y we generally have $b_{X,Y} \neq b_{Y,X}$
- Correlation coefficient is independent of the change of origin and scale. Regression coefficient is independent of change of origin but not of scale.

Types of Regression Analysis:

- Simple Regression (involving two variables where one is independent and the other is dependent variable) and Multiple Regression (involving more than two variables where one variable is dependent and all the other variables are independent)
- Linear Regression ($Y = a + bX$) and Nonlinear Regression (e.g. $Y = a + bX + cX^2$) where X is the independent variable and Y is the dependent variable.
- Partial Regression (to inspect the effect of adding another variable to a model that already has one or more independent variables) and Total Regression (to study the effect of all the important variables on one another).

Examples of Simple Regression and Multiple Regression:

- Example of Simple Regression: Quantity of consumption of food and body weight,
- Example of Multiple Regression: Examination score of a student in an examination depending on various factors like his/her focus while attending the class, his/her preparation level, his/her intelligence, his/her intake of food before the examination, the amount of sleep he/she gets before the examination and his/her attention and strategy during the examination.